

Steps in a Quantum Algorithm

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Week 5 Material

- ▶ This Week - Quantum Gates
- ▶ Reading Assignments are:
 - ▶ 5A - Dancing With Qubits - Chapter 7: ONE QUBIT
 - ▶ 5B- DWQ section 8.1 - “Tensor Products”
- ▶ Outline of Lectures:
 - ▶ 5C - Part 1 - Steps in a Quantum Algorithm
 - ▶ 5D - Part 2 - Unitary, X or Not Gate, and the C-NOT Gate
 - ▶ 5E - Video from Umesh Vazirani about Single Qubit Operations
 - ▶ 5F - Part 3 - Multiple Qubit Gates
 - ▶ 5G - Part 4 - What Gate Choices do we have for our Quantum Program?
 - ▶ 5H - Video from Umesh Vazirani about Multiple Qubit Operations and Tensors
 - ▶ 5I - Part 5 - Recording of Live Lecture to Students about Using IBM Quantum Composer

Fundamentals: Computing

Previous Week's Lecture Summary

- Superposition: is *key for quantum advantage*. The qubit state is $\alpha|0\rangle + \beta|1\rangle$ (a combination of *both* $|0\rangle$ and $|1\rangle$ with complex numbers α and β).
- Entanglement: is *key for quantum advantage*. Separated qubits can exist in states that are correlated in a way that cannot be replicated classically.
- Quantum Measurement: is quantum readout. It *restricts quantum advantage* as we cannot access the c_i . Measurement collapses the state to *either* $|0\rangle$ or $|1\rangle$. Crucially, qubits must be protected from [intended or unintended] measurement.

Basics for quantum computing

- Initialize the quantum state to $|0\rangle$
 - State preparation
- Compute on the quantum states
 - Quantum operations or “gates”
 - State evolution
- Readout the information
 - Measurement

We sometimes call this the
GROUND STATE.

Or think of it as the
Starting State.

Steps in a Quantum Algorithm

From Dancing With Qubits: Chapter 7 - page 228

A quantum algorithm uses several qubits which together constitute a *quantum register*. From the viewpoint of a single qubit, its activity looks like

1. I, the qubit, am in initial state $|0\rangle$.
2. I get moved into a standard superposition. Don't peek!
3. Steps in the algorithm apply zero or more reversible operations to me.
4. I get measured and always become $|0\rangle$ or $|1\rangle$.
5. When these get read out for classical use, they are converted to 0 or 1.

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It's what happens in steps 2 and 3 that make quantum computing interesting. For the unit sphere, which is called the *Bloch sphere* in quantum computing, step 2 means I get moved to a designated point/state on the equator. Step 3 translates to getting moved to other points/states. This movement is a result of what the algorithm is doing and causing to happen.

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The final two steps are like telling the qubit “look, you can't stay somewhere in superposition forever. You need to decide whether you will be 0 or 1. This is not optional.” Where it is in superposition, in what state on the sphere, determines the relative probabilities of the qubit ultimately yielding a classical bit value of 0 or 1.

C² and Measurement

From Dancing With Qubits: Chapter 7 - page 229

We explore and return to this statement throughout the rest of this chapter:

A *qubit*—a *quantum bit*—is the fundamental unit of quantum information. At any given time, it is in a superposition state represented by a linear combination of vectors $|0\rangle$ and $|1\rangle$ in $\mathbb{C}^2$:

$$a|0\rangle + b|1\rangle \quad \text{where} \quad |a|^2 + |b|^2 = 1 .$$

Through *measurement*, a qubit is forced to collapse irreversibly through projection to either $|0\rangle$ or $|1\rangle$. The probability of its doing either is $|a|^2$ and $|b|^2$, respectively. a and b are called *probability amplitudes*.

If necessary, we can convert (“read out”) $|0\rangle$ and $|1\rangle$ to classical bit values of 0 and 1.

Quantum State Representation

From Dancing With Qubits: Chapter 7 - page 235

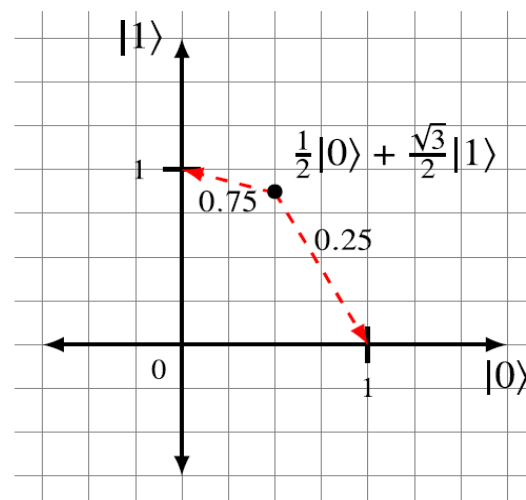
7.3.1 Quantum state representation

If $|\psi\rangle = a|0\rangle + b|1\rangle$ is a quantum state, we know $|\psi|^2 = |a|^2 + |b|^2 = 1$. This is the same as saying $\langle\psi|\psi\rangle = 1$.

The values $|a|^2$ and $|b|^2$ are non-negative numbers in $\mathbb{R}$ that are the probabilities that $|\psi\rangle$ will go to $|0\rangle$ or $|1\rangle$, respectively, when measured.

Consider the following example in which, for illustrative purposes, I have made some simplifications and taken some labeling liberties. Suppose a and b can only be real.

Let $a = \frac{1}{2}$ and $b = \frac{\sqrt{3}}{2}$. Then $|a|^2 = \left|\frac{1}{2}\right|^2 = 0.25$ and $|b|^2 = \left|\frac{\sqrt{3}}{2}\right|^2 = 0.75$. This means that the state of the qubit has a 25% chance of collapsing to $|0\rangle$ when measured and a 75% chance of collapsing to $|1\rangle$.



Bloch Sphere

From Dancing With Qubits: Chapter 7 - page 248

7.5 The Bloch sphere

We describe the state of a qubit by a vector

$$a|0\rangle + b|1\rangle = r_1 e^{i\varphi_1} |0\rangle + r_2 e^{i\varphi_2} |1\rangle$$

in $\mathbb{C}^2$ with r_1 and r_2 non-negative numbers in $\mathbb{R}$.

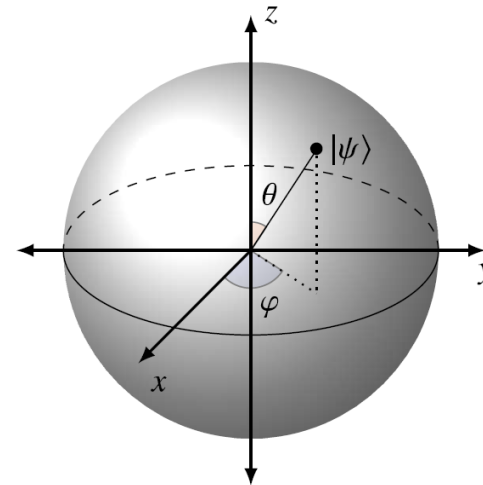
The magnitudes r_1 and r_2 are related by $r_1^2 + r_2^2 = 1$. This is a mathematical condition. We saw in [section 7.3](#) that it's the relative phase of $\varphi_2 - \varphi_1$ that is significant and not the individual phases φ_1 and φ_2 . This is a physical condition and it also means we can take a to be real.

We also saw that we could represent a quantum state as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + \sin\left(\frac{\theta}{2}\right) e^{i\varphi} |1\rangle.$$

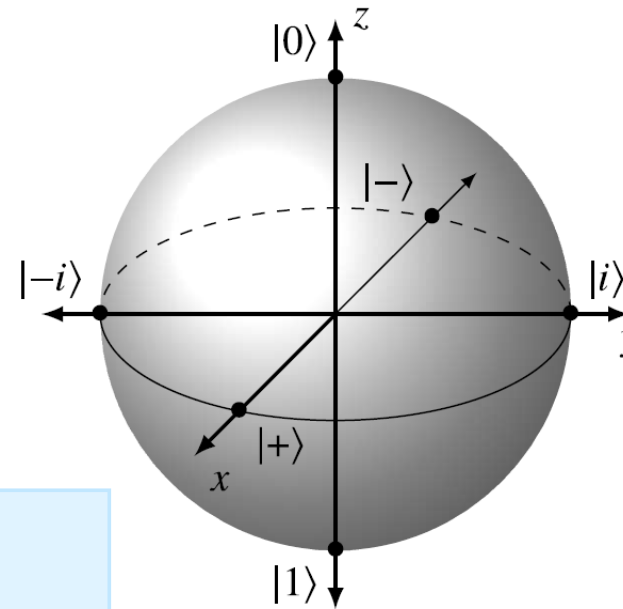
We do this via a non-linear projection and a change of coordinates, and get a point on the surface of the *Bloch sphere*.

The two angles have the ranges $0 \leq \theta \leq \pi$ and $0 \leq \varphi < 2\pi$. θ is measured from the positive z axis and φ from the positive x axis in the xy -plane.



Bloch Sphere (Different Explanation)

From Dancing With Qubits: Chapter 7 - page 252



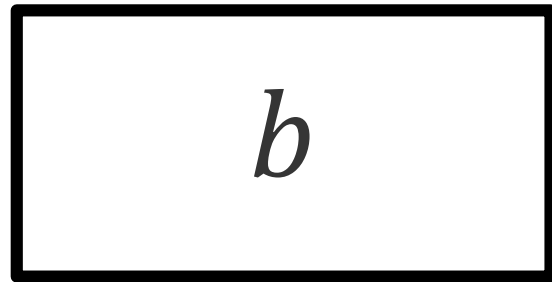
We have now seen three pairs of basis elements and they go by different names:

Basis	Common name	Bloch sphere axis name
$\{ 0\rangle, 1\rangle\}$	Computational	Z
$\{ +\rangle, -\rangle\}$	Hadamard	X
$\{ i\rangle, -i\rangle\}$	Circular	Y

The Y basis is the same as the circular basis. In this case we also have the alternative notation $|i\rangle = |\uparrow\rangle$ and $|-i\rangle = |\downarrow\rangle$.

Unit of information

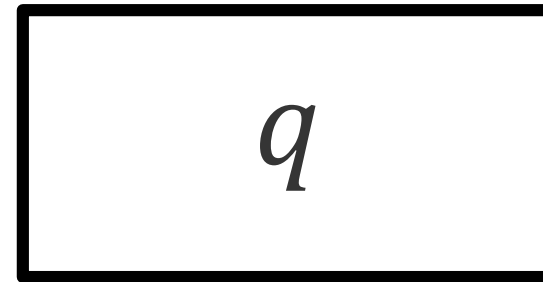
Classical: *bit*



$$b = 0 \text{ or } 1$$

$$b = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \text{"0"}$$

Quantum: *qubit*



$$q = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ or } \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} - \text{"0"}$$

$$\text{with } |c_0|^2 + |c_1|^2 = 1.$$

superposition of 0 and 1 gives quantum advantage

Unit of information

$$q$$

$$q = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ or } \begin{pmatrix} c_0 \\ c_1 \end{pmatrix}$$

with $|c_0|^2 + |c_1|^2 = 1$.

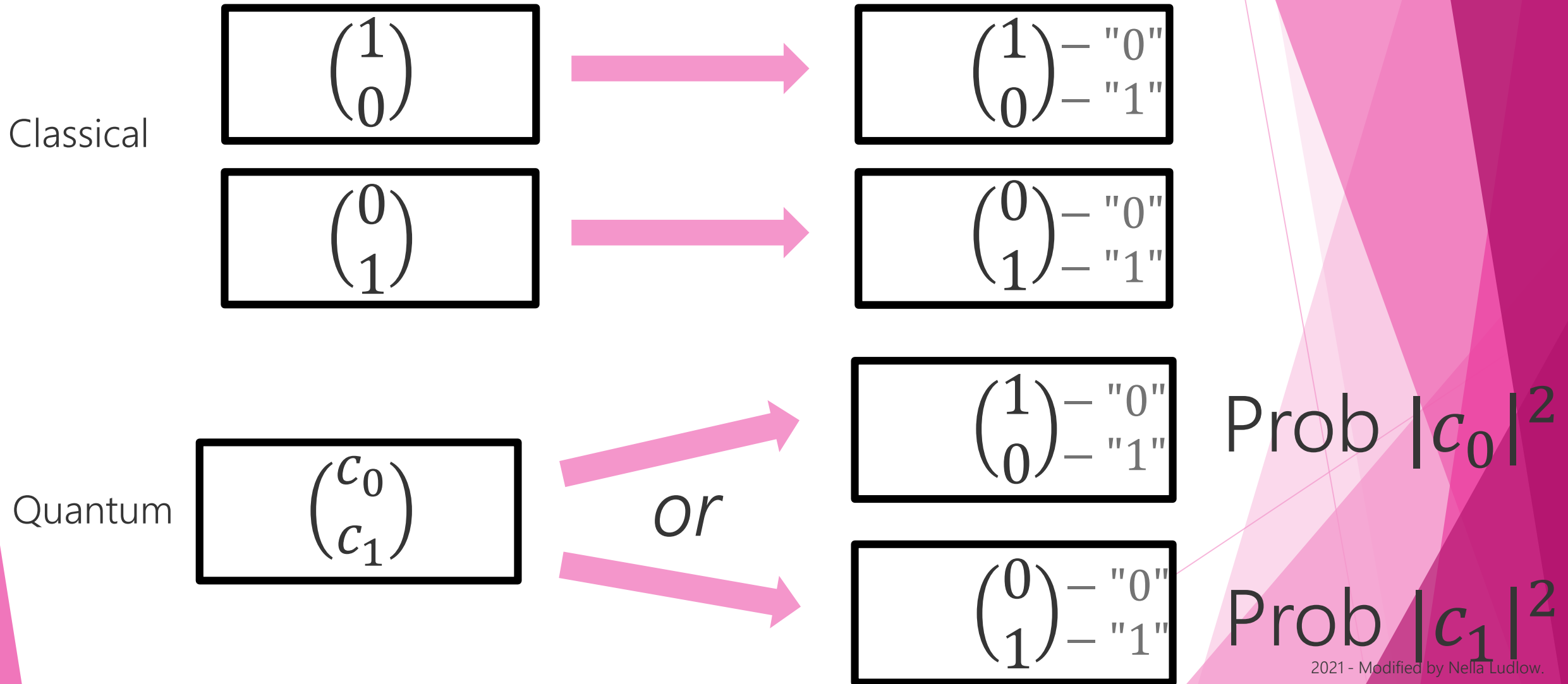
$$c_0, c_1 \text{ are complex: } \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} \in \mathbb{C}^2$$

A qubit is in a linear combination of basis states

$$q = c_0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} c_0 \\ c_1 \end{pmatrix}$$

Basis states "0" and "1"

Readout: quantum measurement



Readout: quantum measurement

- We need measurement for readout.
- Measurement *restricts power of quantum computing*, preventing us from directly learning c_0 and c_1 .
- Measurement destroys superposition.
- A qubit must be protected from unwanted measurement (known as decoherence).

